Lecture Notes

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Lecture 5 found the eigenvalues and eigenvectors of an operator and showed that, with enough independent eigenvectors, its matrix becomes diagonal. This lecture turns that observation into a working theory. We characterize exactly when an operator is *diagonalizable*, prove the *Cayley–Hamilton theorem* (every operator satisfies its own characteristic equation), and use diagonalization to compute *functions of operators*—above all the exponential e^{tA}, which solves linear differential equations and, in the form $e^{-i\hat{H}t/\hbar}$, generates time evolution in quantum mechanics.

The unifying idea is simple: in an eigenbasis an operator is just a list of independent scalings, so anything we can do to numbers—raise to a power, take an exponential, apply a function—we can do to the operator one eigenvalue at a time.

Learning objectives

By the end of this lecture you will be able to:

- Decide diagonalizability and diagonalize a matrix when possible.
- Distinguish algebraic from geometric multiplicity and use the minimal polynomial.
- Compute functions of a diagonalizable operator—powers, e^{tA}, $f(T)$.
- State and apply Cayley–Hamilton and the spectral mapping rule.

1 Diagonalizable operators

Definition 6.1 (Diagonalizable). An operator $T \in \mathcal{L}(V)$ is *diagonalizable* if V has a basis consisting of eigenvectors of T; equivalently, if $\mathcal{M}(T)$ is a diagonal matrix in some basis.

The two formulations agree because the matrix of T in a basis $v_1, \dots, v_n$ is diagonal exactly when $Tv_k = \lambda_k v_k$ for each k, i.e. when every basis vector is an eigenvector. The following theorem collects the practical tests.

Theorem 6.2 (Conditions for diagonalizability). Let $T \in \mathcal{L}(V)$ with $\dim V = n$ and distinct eigenvalues $\lambda_1, \dots, \lambda_m$. The following are equivalent.

1. T is diagonalizable.
2. V has a basis of eigenvectors of T.
3. $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.
4. $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

Proof. (1)$\Leftrightarrow$(2) is the definition. For (2)$\Rightarrow$(4): a basis of eigenvectors splits into vectors lying in the various eigenspaces, so $\sum_k \dim E(\lambda_k, T) \geq n$; the reverse inequality always holds because the eigenspaces form a direct sum (Lecture 5), giving equality. For (4)$\Rightarrow$(3): the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct (Lecture 5), so its dimension is $\sum_k \dim E(\lambda_k, T) = n = \dim V$; a subspace of full dimension is all of V. For (3)$\Rightarrow$(2): concatenating bases of the eigenspaces yields a basis of V consisting of eigenvectors. ■

Corollary 6.3 (Distinct eigenvalues suffice). If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

Proof. The corresponding eigenvectors are independent (Lecture 5), and an independent list of length $\dim V$ is a basis; condition (2) holds. ■

Distinctness is sufficient but not necessary: the identity has the single eigenvalue 1 yet is already diagonal. What fails for a *defective* operator is condition (4)—some eigenspace is too small.

Example 6.4 (A diagonalizable operator). $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has distinct eigenvalues 3, 1 (Lecture 5), so by Corollary 6.3 it is diagonalizable, with eigenvectors $(1, 1), (1, -1)$ forming a basis. ◁

Example 6.5 (A non-diagonalizable operator). The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has only the eigenvalue 1, with $E(1, T) = \operatorname{span}\big((1, 0)\big)$ one-dimensional. The sum of eigenspace dimensions is $1 < 2$, so condition (4) fails and T is not diagonalizable: there is no basis of eigenvectors. ◁

Example 6.6 (A repeated eigenvalue that still diagonalizes). The matrix $A = \left(\begin{smallmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right)$ has eigenvalues 3, 2, 2. The eigenspace $E(2, A)$ solves $(A - 2I)v = 0$, i.e. $v_1 + v_2 = 0$, which is *two*-dimensional: $E(2, A) = \operatorname{span}\big((-1, 1, 0), (0, 0, 1)\big)$. Together with $E(3, A) = \operatorname{span}\big((1, 0, 0)\big)$ the eigenspace dimensions sum to $1 + 2 = 3$, so A is diagonalizable despite the repeated eigenvalue. With $P = \left(\begin{smallmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$,

$$P^{-1}AP = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \operatorname{diag}(3, 2, 2).$$

Contrast the shear of Example 6.5, where the repeated eigenvalue had only a one-dimensional eigenspace. ◁

Example 6.7 (A complete diagonalization). Diagonalize $A = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. The characteristic polynomial $\lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$ gives eigenvalues 5 and 2; solving $(A - 5I)v = 0$ yields the eigenvector $(1, 1)$ and $(A - 2I)v = 0$ yields $(1, -2)$. Assembling these as the columns of P,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad P^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

The recipe—characteristic polynomial, eigenvectors, assemble P—works for every operator possessing a full set of eigenvectors. ◁

2 Diagonalization in practice

Let T on $\mathbb{F}^n$ be diagonalizable, with eigenvectors $v_1, \ldots, v_n$ and eigenvalues $\lambda_1, \ldots, \lambda_n$. Collect the data into

$$P = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \quad \text{(eigenvectors as columns)}, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Since $Av_k = \lambda_k v_k$ reads $AP = PD$ column by column, and P is invertible,

$$A = PDP^{-1}. \tag{6.1}$$

This is the change of basis of Lecture 4, specialized to an eigenbasis (Figure 1). Its great virtue is that powers and, soon, functions collapse to the diagonal:

$$A^k = (PDP^{-1})^k = PD^kP^{-1} = P\, \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)\, P^{-1},$$

the middle factors $P^{-1}P$ telescoping away.

$$\begin{tikzcd} \mathbb{F}^n \arrow[r, "A"] & \mathbb{F}^n \\ \mathbb{F}^n \arrow[u, "P"] \arrow[r, "D = \operatorname{diag}(\lambda_k)"'] & \mathbb{F}^n \arrow[u, "P"'] \end{tikzcd}$$

Figure 1: In the eigenbasis the operator acts as the diagonal D; conjugating by P recovers $A = PDP^{-1}$. Powers and functions are applied to D entrywise.

Example 6.8 (Computing a power). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ with $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $P^{-1} = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, and $D = \operatorname{diag}(3, 1)$,

$$A^k = P \begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

A single diagonalization gives every power at once; computing A^{10} directly would be far more work. ◁

Example 6.9 (Fibonacci numbers by diagonalization). The recurrence $F_{n+1} = F_n + F_{n-1}$ is $\left(\begin{smallmatrix} F_{n+1} \\ F_n \end{smallmatrix}\right) = A\left(\begin{smallmatrix} F_n \\ F_{n-1} \end{smallmatrix}\right)$ with $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so $\left(\begin{smallmatrix} F_{n+1} \\ F_n \end{smallmatrix}\right) = A^n\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$. The eigenvalues of A are the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$, and $A^n = PD^nP^{-1}$ yields Binet's closed form

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Diagonalization turns a recurrence into a formula. ◁

Example 6.10 (Long-run behavior of a Markov chain). The stochastic matrix $A = \left(\begin{smallmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{smallmatrix}\right)$ has eigenvalues 1 and 0.5. Expanding any initial state in the eigenbasis, A^n leaves the $\lambda = 1$ component fixed and shrinks the $\lambda = 0.5$ component by $0.5^n \to 0$. Hence $A^n x_0$ converges to the $\lambda = 1$ eigenvector—the stationary distribution $(0.6, 0.4)$—for any initial probability distribution x_0. Diagonalization exposes the long-run behavior immediately. ◁

Remark 6.11 (Decoupling). Physically, passing to the eigenbasis is *decoupling*. A system of coupled linear differential equations $\dot{x} = Ax$ becomes, in eigen-coordinates $y = P^{-1}x$, the uncoupled set $\dot{y}_k = \lambda_k y_k$—the normal modes of a vibrating system, each oscillating independently.

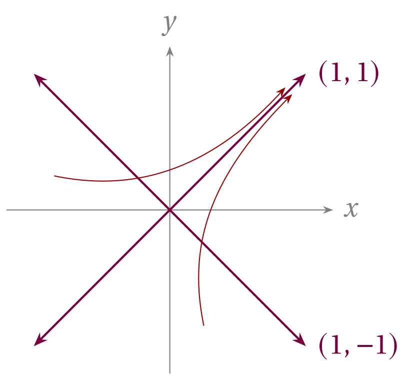

Figure 2: Phase portrait of $\dot{x} = Ax$ for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ (eigenvalues 3, 1): trajectories flow outward and bend toward the dominant eigendirection $(1, 1)$.

Example 6.12 (Normal modes of two coupled masses). Two equal masses joined by identical springs obey $\ddot{x} = -Kx$ with $K = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ in suitable units. The eigenvalues of K are 1 and 3, with eigenvectors $(1,1)$ and $(1,-1)$. In these *normal coordinates* the system decouples into two independent oscillators: the in-phase mode $(1,1)$ at angular frequency $\omega = 1$ and the out-of-phase mode $(1,-1)$ at $\omega = \sqrt{3}$. Diagonalizing K *is* finding the normal modes. ◁

3 The Cayley–Hamilton theorem

A striking fact ties an operator to its characteristic polynomial $\chi_T(\lambda) = \det(\lambda I - T)$.

Theorem 6.13 (Cayley–Hamilton). Every operator T on a finite-dimensional complex vector space satisfies its own characteristic equation: $\chi_T(T) = 0$.

Proof for diagonalizable T. Write $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. For any polynomial p, $p(A) = P\,p(D)\,P^{-1}$ and $p(D) = \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n))$. Taking $p = \chi_A$ and using that each λ_k is a root of χ_A,

$$\chi_A(A) = P\ \operatorname{diag}\big(\chi_A(\lambda_1), \dots, \chi_A(\lambda_n)\big)P^{-1} = P\ \operatorname{diag}(0, \dots, 0)\,P^{-1} = 0. \qquad ■$$

Proof in general. Over $\mathbb{C}$ every operator has an upper-triangular matrix (Lecture 5, where triangularizability is sketched) with diagonal entries $\lambda_1, \dots, \lambda_n$, and then $\chi_T(\lambda) = (\lambda - \lambda_1)\cdots(\lambda - \lambda_n)$. The triangular-product lemma of Lecture 5 gives $(T - \lambda_1 I)\cdots(T - \lambda_n I) = 0$ for such an operator, and that product is exactly $\chi_T(T)$, so $\chi_T(T) = 0$. ■

Corollary 6.14 (Inverses and powers as low-degree polynomials). If T is invertible with $\chi_T(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$ (so $c_0 = (-1)^n \det T \neq 0$), then

$$T^{-1} = -\tfrac{1}{c_0}\big(T^{n-1} + c_{n-1}T^{n-2} + \cdots + c_1 I\big).$$

More generally every power T^k with $k \geq n$ reduces to a polynomial in T of degree $< n$.

Proof. Cayley–Hamilton gives $T^n + \cdots + c_1 T + c_0 I = 0$. Solve for $c_0 I$ and multiply by $T^{-1}/(-c_0)$ for the first claim; rearranging the same relation expresses T^n (hence inductively any higher power) through lower ones. ■

Example 6.15 (Inverse from Cayley–Hamilton). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$, so $A^2 - 4A + 3I = 0$ and $A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, which one checks satisfies $AA^{-1} = I$. ◁

Example 6.16 (Cayley–Hamilton, verified by hand). For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$, $\chi_A(\lambda) = \lambda^2 - 5\lambda - 2$, and indeed

$$A^2 - 5A - 2I = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ 15 & 20 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The operator annihilates its own characteristic polynomial. ◁

Example 6.17 (Reducing a high power). From $A^2 = 5A + 2I$ in Example 6.16, every higher power collapses to a combination of A and I: $A^3 = A \cdot A^2 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I$, and inductively $A^k = \alpha_k A + \beta_k I$ for scalars α_k, β_k. Cayley–Hamilton replaces unbounded powers with arithmetic in the two “coordinates” I and A. ◁

Remark 6.18 (Minimal polynomial). The monic polynomial of least degree annihilating T is its *minimal polynomial*; it divides χ_T and shares its roots (the eigenvalues). An operator is diagonalizable exactly when its minimal polynomial has no repeated roots. For the shear

$\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ the minimal polynomial is $(\lambda-1)^2$—since $A-I\neq 0$ but $(A-I)^2=0$—and its repeated root flags the non-diagonalizability we found in Example 6.5; whenever the eigenvalues are distinct the minimal polynomial is $\prod_k(\lambda-\lambda_k)$ with simple roots, confirming Corollary 6.3.

4 Functions of operators

We have already used *polynomials* of an operator. For a diagonalizable T we can apply *any* function defined on the eigenvalues.

Definition 6.19 (Function of a diagonalizable operator). If $T = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1,\dots,\lambda_n)$ and f is defined at each λ_k, set

$$f(T) = P\,f(D)\,P^{-1}, \qquad f(D) = \operatorname{diag}\big(f(\lambda_1),\dots,f(\lambda_n)\big).$$

This agrees with the power-series definition whenever the series converges, and is independent of the choice of eigenbasis.

The most important example is the *exponential*, defined for every operator by the everywhere-convergent series

$$e^A = \sum_{k=0}^{\infty}\frac{A^k}{k!}.$$

For diagonalizable $A = PDP^{-1}$ this sums to $e^A = P\operatorname{diag}(e^{\lambda_1},\dots,e^{\lambda_n})\,P^{-1}$. Its defining property is the one that matters for dynamics.

Proposition 6.20 (Exponential and linear flow). For fixed A, the function $t\mapsto e^{tA}$ satisfies $\dfrac{d}{dt}e^{tA} = A\,e^{tA}$ and $e^{0A}=I$. Hence $x(t)=e^{tA}x_0$ is the unique solution of $\dot{x}=Ax$ with $x(0)=x_0$.

Proof. Differentiate the series term by term (it converges uniformly on bounded t-intervals): $\frac{d}{dt}\sum_k\frac{t^kA^k}{k!}=\sum_{k\geq 1}\frac{t^{k-1}A^k}{(k-1)!}=A\,e^{tA}$. Then $\frac{d}{dt}\big(e^{tA}x_0\big)=Ae^{tA}x_0$, and $x(0)=x_0$. ■

Example 6.21 (Exponential of a 2×2 via the eigenbasis). With $A=\left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)=PDP^{-1}$, $D=\operatorname{diag}(3,1)$ as before,

$$e^{tA} = P\begin{pmatrix}e^{3t}&0\\0&e^{t}\end{pmatrix}P^{-1} = \frac{1}{2}\begin{pmatrix}e^{3t}+e^{t}&e^{3t}-e^{t}\\e^{3t}-e^{t}&e^{3t}+e^{t}\end{pmatrix}.$$

The solution of $\dot{x}=Ax$ is a combination of the modes e^{3t} and e^{t} along the eigendirections $(1,1)$ and $(1,-1)$. ◁

Example 6.22 (Solving an initial value problem). For the same A with $x(0)=(1,0)$, expand the initial data in the eigenbasis, $x(0)=\frac{1}{2}(1,1)+\frac{1}{2}(1,-1)$, evolve each mode by its own exponential, and recombine:

$$x(t)=\tfrac{1}{2}e^{3t}(1,1)+\tfrac{1}{2}e^{t}(1,-1)=\tfrac{1}{2}\big(e^{3t}+e^{t},\ e^{3t}-e^{t}\big).$$

The dominant mode $e^{3t}(1,1)$ dictates the long-time behavior, and the answer agrees with the first column of e^{tA} found above. ◁

Example 6.23 (Spin rotation: $e^{i\theta\sigma_x}$). Because $\sigma_x^2=I$, the exponential series splits into even and odd powers:

$$e^{i\theta\sigma_x}=\sum_j\frac{(i\theta)^{2j}}{(2j)!}I+\sum_j\frac{(i\theta)^{2j+1}}{(2j+1)!}\sigma_x=\cos\theta\,I+i\sin\theta\,\sigma_x.$$

This is the operator that rotates a spin about the x-axis; the same calculation with any operator squaring to I produces a cos/sin pair. It is the two-level version of the evolution $e^{-i\hat{H}t/\hbar}$. ◁

Example 6.24 (A square root of an operator). Since $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right) = PDP^{-1}$ with $D = \operatorname{diag}(3, 1)$ and positive eigenvalues, it has a square root

$$\sqrt{A} = P \ \operatorname{diag}(\sqrt{3}, 1)\, P^{-1} = \frac{1}{2}\begin{pmatrix} \sqrt{3}+1 & \sqrt{3}-1 \\ \sqrt{3}-1 & \sqrt{3}+1 \end{pmatrix},$$

which satisfies $(\sqrt{A})^2 = A$. A function of an operator is no harder than the same function of its eigenvalues. ◁

Remark 6.25 (Identities transport to operators). Any scalar identity among functions holds for diagonalizable operators, since both sides act eigenvalue by eigenvalue. For instance $\sin^2 A + \cos^2 A = I$ and $e^A e^{-A} = I$ hold for every diagonalizable A (and, by a continuity argument, for all A).

Proposition 6.26 (Spectral decomposition). If T is diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_m$, there are projections $P_1, \dots, P_m$ (each onto an eigenspace along the others) with

$$P_i P_j = 0 \ \ (i \neq j), \qquad P_1 + \cdots + P_m = I, \qquad T = \lambda_1 P_1 + \cdots + \lambda_m P_m,$$

and consequently $f(T) = \sum_k f(\lambda_k) P_k$ for any f defined on $\operatorname{spec}(T)$.

Proof. In the eigenbasis T is constant ($= \lambda_k$) on each eigenspace block; let P_k be the projection picking out the $E(\lambda_k, T)$ block. These satisfy the stated relations, $T = \sum_k \lambda_k P_k$, and applying f blockwise gives $f(T) = \sum_k f(\lambda_k) P_k$. ■

This is the finite-dimensional shadow of the spectral theorem of Lecture 10, in which the P_k become *orthogonal* projections.

Example 6.27 (Spectral decomposition in coordinates). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ the eigenprojections onto $E(3) = \operatorname{span}(1, 1)$ and $E(1) = \operatorname{span}(1, -1)$ are

$$P_1 = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P_2 = \tfrac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

with $P_1 + P_2 = I$ and $P_1 P_2 = 0$. Then $A = 3P_1 + P_2$, and Proposition 6.26 gives $A^k = 3^k P_1 + P_2$ and $e^A = e^3 P_1 + e P_2$ with no matrix multiplication at all—recovering Example 6.8 instantly. ◁

5 Spectral mapping and time evolution

Functions act on the spectrum in the obvious way.

Proposition 6.28 (Spectral mapping). If T is diagonalizable and f is defined on $\operatorname{spec}(T)$, then $\operatorname{spec}\big(f(T)\big) = \{f(\lambda) : \lambda \in \operatorname{spec}(T)\}$, and every eigenvector of T for λ is an eigenvector of $f(T)$ for $f(\lambda)$.

Proof. If $Tv = \lambda v$ then $f(T)v = Pf(D)P^{-1}v$ acts on the eigenvector v by the scalar $f(\lambda)$, since in eigen-coordinates v is a standard basis vector and $f(D)$ scales it by $f(\lambda)$. ■

The converse inclusion for eigenvectors fails when f is not injective on the spectrum: for $T = \mathrm{diag}(1, -1)$ and $f(\lambda) = \lambda^2$ we get $f(T) = I$, every nonzero vector of which is an eigenvector, while T has only two eigendirections. Distinct eigenvalues can be merged by f, and merging enlarges an eigenspace.

Example 6.29 (Spectrum of a function). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ with spectrum $\{1, 3\}$, the operator e^A has spectrum $\{e, e^3\}$ and A^{10} has spectrum $\{1, 3^{10}\}$, all with the same eigenvectors $(1, -1)$ and $(1, 1)$. Once the eigenvectors are known, no function of A requires recomputing them. ◃

Example 6.30 (Oscillation from imaginary eigenvalues). The generator $A = \left(\begin{smallmatrix} 0 & -\omega \\ \omega & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm i\omega$, and a short computation (or the cos/sin split, since $A^2 = -\omega^2 I$) gives

$$e^{tA} = \begin{pmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{pmatrix}.$$

So $\dot{x} = Ax$ is rotation at angular frequency ω: imaginary eigenvalues produce oscillation, exactly as real ones produced the growth and decay of Example 6.21. This is the linear-algebra heart of simple harmonic motion. ◃

Remark 6.31 (Quantum time evolution). A quantum state evolves in time by $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$. In an energy eigenbasis $\hat{H} = \mathrm{diag}(E_1, \dots, E_n)$, so

$$e^{-i\hat{H}t/\hbar} = \mathrm{diag}\big(e^{-iE_1 t/\hbar}, \dots, e^{-iE_n t/\hbar}\big) :$$

each stationary state merely acquires a phase $e^{-iE_n t/\hbar}$, and a general state is a superposition of these independently rotating phases. Diagonalizing the Hamiltonian *is* solving the dynamics—the payoff that the spectral theorem (Lecture 10) guarantees for every observable.

Remark 6.32 (Simultaneous diagonalization). Commuting diagonalizable operators can be brought to diagonal form in a *single* basis—the common eigenvectors of Lecture 5. Physically, compatible observables share an eigenbasis, so a system can hold simultaneously definite values for both, in sharp contrast with incompatible observables such as σ_x and σ_z, which share no eigenbasis.

6 When diagonalization fails

Not every operator is diagonalizable, but the generalized eigenvectors of Lecture 5 restore order. The obstruction is a *nilpotent* operator—one with $N^k = 0$ for some k—the part of an operator that no change of basis can diagonalize away. On a single defective eigenvalue $A = \lambda I + N$ with N nilpotent, and since λI and N commute the exponential series *terminates*:

$$e^{tA} = e^{\lambda t}\, e^{tN} = e^{\lambda t}\Big(I + tN + \tfrac{t^2}{2!}N^2 + \cdots + \tfrac{t^{k-1}}{(k-1)!}N^{k-1}\Big).$$

For the defective shear block $A = \left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$, the nilpotent part $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $N^2 = 0$, so

$$e^{tA} = e^{\lambda t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The extra factor t is the signature of a defective eigenvalue: it produces solutions $t\, e^{\lambda t}$, the same ones that appear for repeated roots of a linear differential equation.

> **Theorem 6.33 (Jordan normal form · cited).** Every operator on a finite-dimensional complex space has a basis of generalized eigenvectors in which its matrix is block diagonal,

each block a *Jordan block*

$$J = \lambda I + N, \qquad N = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix},$$

carrying the eigenvalue λ on the diagonal and the nilpotent shift N on the superdiagonal. Each block is one Jordan chain of generalized eigenvectors (Lecture 5).

The shift N sends each generalized eigenvector to the one below it, $e_j \mapsto e_{j-1} \mapsto \cdots \mapsto e_1 \mapsto 0$—a ladder that descends to zero in finitely many steps, which is exactly what nilpotency means.

Theorem 6.34 (Jordan–Chevalley decomposition • cited). On a complex vector space of finite dimension, every operator A splits uniquely as

$$A = D + N, \qquad DN = ND,$$

with D diagonalizable and N nilpotent, and both D and N polynomials in A.

Because D and N are polynomials in A, they commute with everything A commutes with, so the splitting respects any symmetry A has—this is what makes it more than a normal-form trick. It is also the cleanest route to $e^{tA} = e^{tD}e^{tN}$ for an arbitrary matrix: the diagonal factor supplies the oscillation and decay, the nilpotent factor the polynomial-in-t corrections.

Remark 6.35 (Nilpotents and ladder operators). Nilpotent operators are the ladder operators of physics, seen in finite dimensions. On a spin-j representation (dimension $2j+1$) the raising operator J_+ steps $|m\rangle \mapsto |m+1\rangle$ until it annihilates the top state, so $J_+^{2j+1} = 0$: J_+ is nilpotent, and its action is precisely the shift N of a Jordan block. The harmonic-oscillator lowering operator truncated to N levels is nilpotent in the same way ($a^N = 0$), and the oscillator ladder of Lectures 13–14 is its infinite-dimensional limit. Nilpotency squared to zero, $Q^2 = 0$, is also the defining relation of the supercharge in supersymmetric quantum mechanics, of the BRST charge, and of the exterior derivative—the algebra behind cohomology.

Remark 6.36 (Where defective operators are physical, and a note on self-study). Self-adjointness will rule defectiveness out entirely (Lectures 9–10), so the observables of quantum mechanics are always diagonalizable. Defective operators belong instead to *non-Hermitian* physics: the critically damped oscillator, whose repeated root gives the tell-tale $t\,e^{-\gamma t}$ above, and the *exceptional points* of PT-symmetric and open systems, where two eigenvectors coalesce and the Hamiltonian becomes a genuine Jordan block—a measurable effect in optics and cavity physics. For a *self-adjoint* family $H_0 + \varepsilon V$ no such coalescence can occur at real ε; it can at complex ε, and how near those points come to the real axis is exactly what limits the perturbation series of Lecture 10. The Jordan and Jordan–Chevalley results are offered here for self-study: we state their structure without proving the construction, and carry forward only the principle—diagonal where possible, a nilpotent correction where not.

Summary of main concepts

An operator is diagonalizable iff it has a basis of eigenvectors, equivalently iff its eigenspace dimensions sum to $\dim V$; having $\dim V$ distinct eigenvalues is a convenient sufficient condition. In an eigenbasis $A = PDP^{-1}$, so powers and functions are computed one eigenvalue at a time: $A^k = PD^kP^{-1}$ and $f(T) = Pf(D)P^{-1}$, with the exponential e^{tA} solving $\dot{x} = Ax$. Every operator satisfies its own characteristic equation $\chi_T(T) = 0$ (Cayley–Hamilton), which expresses inverses and high powers as low-degree polynomials in T. Spectral mapping sends eigenvalues λ to $f(\lambda)$; in quantum mechanics this is exactly why diagonalizing the Hamiltonian solves the time evolution $e^{-i\hat{H}t/\hbar}$.

- **Diagonalizable**—has an eigenbasis $\iff$ eigenspace dimensions sum to $\dim V$; $\dim V$ distinct eigenvalues suffice.
- **Diagonalized form**—$A = PDP^{-1}$, so powers and functions act one eigenvalue at a time; e^{tA} solves $\dot{x} = Ax$.
- **Cayley–Hamilton**—$\chi_T(T) = 0$; expresses inverses and high powers as low-degree polynomials in T.
- **Spectral mapping**—f sends eigenvalue λ to $f(\lambda)$—why diagonalizing H solves $e^{-iHt/\hbar}$.
- **Jordan form**—the canonical form when diagonalization fails (self-study).

Exercises for this lecture are in the companion sheet `exercises-6.pdf`.